

Secure File Storage System with AES-256 Report

1. Introduction

Data security is an important aspect of modern computing, especially when storing sensitive files on local systems. Unauthorized access, data theft, and file tampering are common security threats. Encryption is one of the most effective ways to protect files from unauthorized access. This project focuses on developing a Secure File Storage System using AES-256 encryption to protect files stored on a local system. The system ensures confidentiality and integrity of files by encrypting and decrypting files securely.

2. Abstract

This project presents a Secure File Storage System developed using Python that uses AES-256 encryption for secure file storage. The system allows users to upload files, encrypt them, and store them with a .enc extension. It also stores metadata such as file name, timestamp, and hash value securely. The system supports secure file retrieval through decryption and verifies file integrity using hash verification to prevent tampering. This project helps in understanding encryption, data integrity, and secure file storage techniques.

3. Tools Used

Python, Cryptography Library (Fernet/AES), PyQt5 or Command Line Interface (CLI), Hashlib for hash verification, File Handling System.

4. Steps Involved in Building the Project

1. Install Python and required libraries such as cryptography and PyQt5. 2. Generate a secure AES-256 encryption key. 3. Create a function to encrypt files and save them with a .enc extension. 4. Store file metadata such as file name, timestamp, and hash value securely. 5. Create a function to decrypt files when authorized users request access. 6. Implement hash verification to ensure file integrity and detect tampering. 7. Design a PyQt5 GUI or CLI interface for file upload, encryption, and decryption. 8. Test the system with different file types. 9. Store encrypted files securely in a local directory.

5. Conclusion

The Secure File Storage System using AES-256 successfully encrypts and decrypts files while ensuring data confidentiality and integrity. The system protects files from unauthorized access and detects file tampering using hash verification. This project improves understanding of cryptography, encryption algorithms, and secure storage systems. The system can be further enhanced by adding user authentication and cloud storage integration.